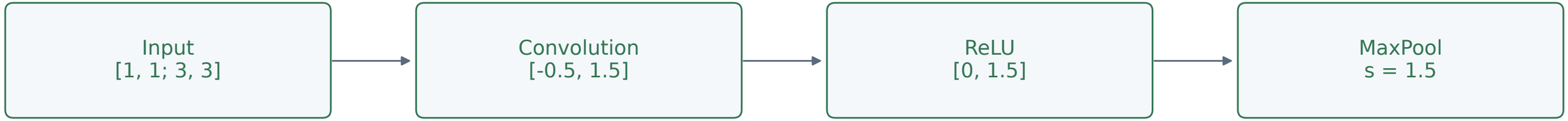
03 | Original CNN endpoint logic: current gates and winner

Analytic CNN: shared [1/2, 1/2] convolution, bias -1.5, ReLU, then max pooling.



Endpoint pooled unit

0.000	0.000
1.500	1.500

$\Sigma = 3.0000$

Same CNN, Path IG

0.000	0.000
0.750	0.750

$\Sigma = 1.5000$

$$s(\alpha x) = \max(3\alpha - 1.5, 0)$$

$$\nabla s(x) = P_{2,:}$$

$$L^{end} = x \odot P_{2,:}$$

$$IG = x \odot \int_{1/2}^{1} P_{2,:}\, d\alpha$$

Endpoint sum = 3; response = 1.5. Bias sets the gate threshold but receives no direct input seed.

The pooled winner is unique at x. Along the path, the output is zero until the ReLU threshold is crossed.

This is a paper-formula CNN example, not a reproduction of the paper's trained-model images.